Ethan Ransom

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Professional Summary

First Class Honours Computer Science graduate from Swansea University with a BTEC Level 3 Extended Diploma in Computing. I am applying for the Digital (Digital & Data) Analyst role to utilise the technical knowledge I gained from my university education and develop it into industrial expertise. My dissertation applied machine learning techniques to medical imaging combining complex medical research with scripting/programming skills. I am detail-oriented, I enjoy deconstructing complex problems into interpretable terms, and working with and manipulating data in novel and interesting ways.

Education

Swansea University, Swansea June 2026

B.Sc. Computer Science, First Class Honours (72.83% average)

- Highest marks: Web Application Development (96%), Big Data and Machine Learning (88%), Critical Systems (81%), Visual Computing (73%) and Intelligent Robotics (73%).
- Final-year dissertation and project modules: 75% and 79%.
- Other coursework: Brain-Inspired Artificial Intelligence, Software Engineering, Database Systems, Algorithms.

City of Bristol College, Bristol June 2022

BTEC Level 3 National Extended Diploma in Computing (IT), DDD*

Gordano School, Portishead June 2020

GCSEs: Combined Science 6-6; Computer Science 5; Geography 5; French 5; Mathematics 4; English Language / English Literature 4;

Projects

Machine Learning Segmentation and Classification of Prostate Cancer MRIs 2025–2026

Final Year Dissertation (First-Class)

- Adapted nnU-Net's preprocessing and data pipeline so SwinUNETR and U-Net++ could be trained from scratch and evaluated on the same inputs, using Nibabel and NumPy to handle the 3D bi-parametric MRI volumes (1,500 patient cases) and matching nnU-Net's per-channel z-score normalisation for a like-for-like comparison.
- Compared how a hybrid Vision Transformer (SwinUNETR) and a deeply supervised nested CNN (U-Net++) differed in localising and diagnosing lesions.
- Built a Streamlit dashboard with a 3D MRI slice viewer, interactive ROC curves and confusion matrices, so clinical stakeholders could explore the results without reading code.

Experience

Waiter, Tiffin At The Beach, Clevedon, Somerset June–October 2024

- Worked busy service periods, keeping on top of several tables and orders at once while staying calm.
- Coordinated closely with kitchen and front-of-house staff to keep service running smoothly.

Skills

- **Programming languages:** Python, Java, C++, C, SQL, PHP, HTML, CSS, Spark, Ada, Laravel
- **Computing knowledge:** Algorithms, Databases, Software Engineering, Web Development, Critical Systems, Automata and Formal Language Theory
- **ML and AI:** Deep learning, CNNs, Vision Transformers, Image Segmentation and Classification, Computer Vision, Model Benchmarking and Evaluation, MONAI, SwinUNETR, U-Net/U-Net++, nnU-Net
- **Medical imaging:** Medical Image Analysis and Segmentation, Multiparametric/Bi-parametric MRI, DICOM/NIfTI, Nibabel
- **Data and visualisation:** NumPy, Streamlit, Data Preprocessing, Clinical Data Analytics.
- **Engineering:** Git/GitHub, Docker, REST APIs, Unit Testing, System Design.
- **Communication:** Presenting technical results clearly to non-technical audiences.